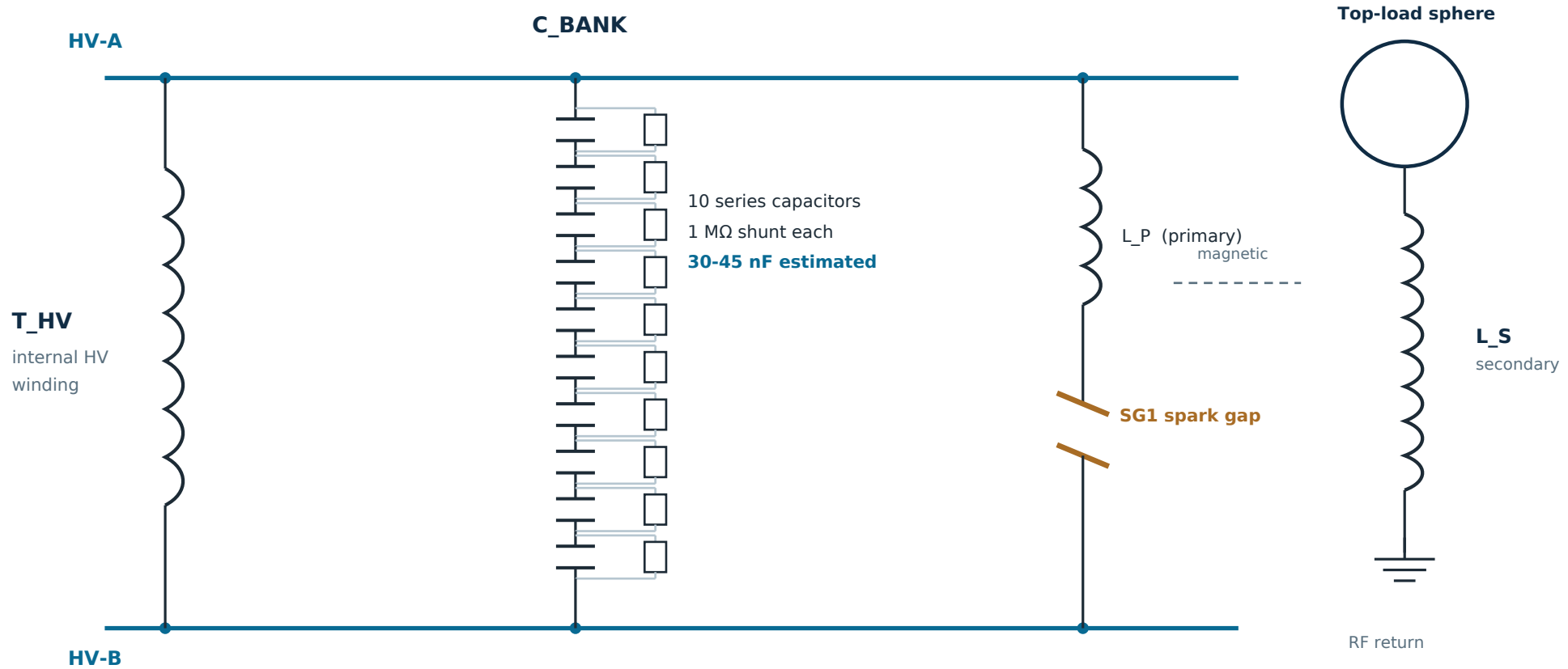
High-voltage network - reconstructed topology

Based on your wiring sketch, capacitor-bank photo and kit assembly reference

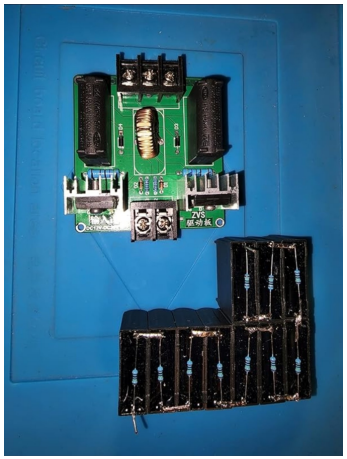


Topology supported: T_HV and C_BANK span HV-A/HV-B; spark gap + L_P form a second branch.

L_S has no metallic connection to L_P; its bottom return needs deliberate RF grounding.

ZVS driver - photographed hardware

Confirmed hardware and probable Mazzilli/Royer-family circuit behaviour; PCB netlist still unverified



Visible on green PCB

ZVS board silkscreen confirms driver type.

Two heatsinked MOSFET positions.

Two large blue/black film capacitors.

Two fast diode positions and gate parts.

Central toroidal power choke.

Two-pin DC input; 3-pin transformer output.

Working block connections

DC input

12-36 V nominal

ZVS driver

2 MOSFETs + LC tank

Flyback HV transformer

Center-tapped LV winding

Likely topology: center tap fed from a choke; winding ends switched by MOSFET drains.

UNVERIFIED: pin assignments, resonant C1/C2 net connection, gate-drive diode orientation.

Important: the two driver tank capacitors are NOT members of the ten-capacitor HV bank.

Component values and resonance calculations

Specified values versus those visible in your photographs, with assumptions explicitly separated

Item	Evidence	Engineering value
HV capacitor bank	10 series units + 1 MΩ shunts	30-45 nF nominal
Driver film capacitor	0.32 µF printing on one cap	2 installed; exact connection unknown
Transformer drive	Three wires at added winding	Center tap + two outer ends
Primary turn count	Short copper winding on white support	Not reliably countable
Secondary winding	Copper solenoid; Ø ~40 mm, h ~247 mm	L_S and self-C unknown
Top-load sphere	Reference photo Ø ~60 mm	Isolated sphere ~3.34 pF

Primary resonant frequency (illustrative)

$$f_P = 1 / [2\pi \sqrt{(L_P C_{BANK})}]$$

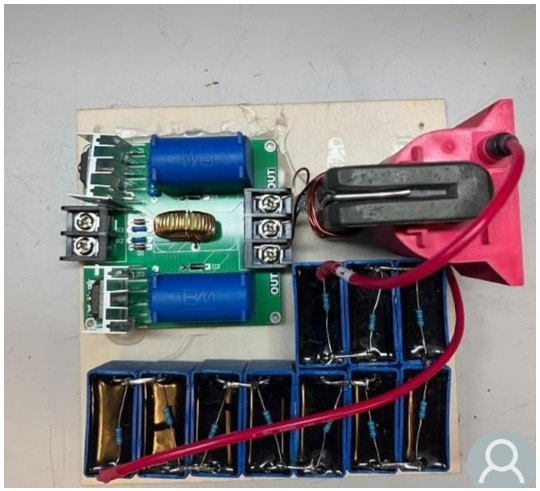
Using 32 nF ONLY if the ten HV capacitors each measure 0.32 µF:

Assumed L_P	f (32 nF)	30-45 nF range	Interpretation
0.30 µH	1.624 MHz	1.370-1.678 MHz	Estimate only
0.50 µH	1.258 MHz	1.061-1.299 MHz	Estimate only
0.75 µH	1.027 MHz	0.866-1.061 MHz	Estimate only
1.00 µH	0.890 MHz	0.750-0.919 MHz	Estimate only

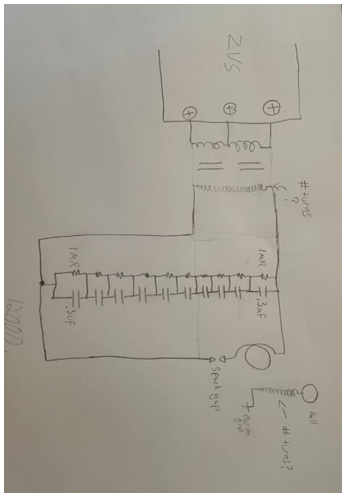
A voltage, frequency, discharge energy, coupling factor, or true L_P cannot be inferred from the photos alone.

Photographic evidence and remaining uncertainty

What is now independently visible - and what would still need continuity tracing to certify a netlist



Assembled ZVS, flyback and HV bank



Hand-drawn net diagram



Operating streamer

Verified (from the supplied evidence)

1. Capacitor-bank branch is connected across transformer HV output.
2. The spark-gap/primary-winding branch is across the same nodes.
3. ZVS output drives a center-tapped, user-wound transformer winding.
4. Coil produces multiple streamer branches during operation.

Still unverified

1. Exact transistor gate/drain/source netlist.
2. Actual HV-bank capacitor values.
3. Transformer winding polarities and turns.
4. Actual L_P, L_S, resonant frequencies.
5. Verified secondary RF return connection.

Safety: disconnect the supply; HV capacitors can retain lethal charge despite shunt resistors. No live probing of exposed HV.

Source basis: supplied photos/sketch; PEMENOL GY19206 kit guide (26-page installation manual and 10-page summary).